

Critical Review: BEV Traction Motor Material Composition — Source Cross-Validation and Value Confirmation

RAWCLIC Project Critical Review / September 2026

1. Review mandate, method and status framework

1.1 Scope of the audit

✓ CONFIRMED — This review audits the material, mass, power-density, market-adoption, technology-development and regulatory claims in the original “Practical BEV Traction-Motor Architecture, Weight and Materials Outlook.” The original report is treated only as the claim under review, not as independent corroboration [1].

✓ CONFIRMED — The associated RAWCLIC Deliverable 3.1 models BEV traction-motor composition for permanent-magnet motors, induction motors and electrically excited synchronous machines. Its numerical dataset is intended for vehicle-segment stock-and-flow modelling, and component weights are reported at vehicle level rather than necessarily per individual motor [2].

🔧 ADJUSTED — The central audit question is therefore not whether an original value is plausible, but whether independent evidence confirms the same topology, power rating, duty rating, component boundary and number of motors per vehicle. Values failing any of these boundary tests are retained only as planning assumptions or marked as gaps.

1.2 Systematic cross-validation procedure

✓ CONFIRMED — The review applies five sequential tests:

1. **Claim identification:** capture the original Min | Mode | Max and determine whether it concerns a measured product, generalized motor class or constructed scenario [1].
2. **Source independence:** exclude the original report and RAWCLIC workbook as independent confirmation of their own values [2] [1].
3. **Boundary matching:** distinguish motor-only from inverter, gearbox, differential, cooling circuit, retained fluid, e-axle and complete vehicle-level propulsion boundaries.
4. **Rating matching:** distinguish peak or short-duration power from continuous thermally sustainable power.
5. **Triangulation:** require two or three independent eligible sources for every confirmed or adjusted numerical value.

🔧 ADJUSTED — A source does not become independent merely because the same supplier claim is repeated by automotive media. Supplier specifications and reports quoting those specifications

remain one originating evidence stream unless separate testing or peer-reviewed characterization is available.

1.3 Reliability grading

Evidence class	Reliability	Audit treatment	Status
Peer-reviewed literature, EU deliverables and standards	100%	Highest evidentiary weight; still requires matching system and duty boundaries	 CONFIRMED
Industry analysts, national laboratories, JRC and Fraunhofer	80%	Strong directional or benchmark evidence; proprietary forecast methods remain a limitation	 CONFIRMED
OEM or supplier press, white papers and established technical automotive journalism	60%	Suitable for topology and disclosed specifications; performance remains company-attributed without independent testing	 ADJUSTED
Recognised engineering news and conference presentations	40%	Corroborative only, especially for prototypes and claimed records	 ADJUSTED

 **CONFIRMED** — Sources below 40% are excluded. Wikipedia, Reddit, personal blogs, general market-report aggregators, search snippets and ResearchGate copies are not used as evidence.

1.4 Source-boundary controls

Boundary issue	Required audit treatment	Status
Motor-only versus e-axle	Do not use an e-axle or drive-unit mass as motor mass unless separable subassembly masses are disclosed	 CONFIRMED
Peak versus continuous power	Report the relevant rating explicitly; do not divide peak power by mass and present the result as continuous density	 CONFIRMED
One motor versus vehicle total	Preserve both per-motor and per-vehicle values; never normalize a dual-motor vehicle to one motor without disclosure	 CONFIRMED
Housing and cooling	State whether the housing, end shields, coolant jacket, pump, hoses and retained fluid are included	 CONFIRMED
Axial versus radial flux	Treat flux direction separately from magnet type; most axial-flux traction machines reviewed here are permanent-magnet machines	 CONFIRMED
Installed versus marketed share	Do not merge vehicle-model availability, vehicle registrations and installed-motor counts	 CONFIRMED
Nominal "150 kW class"	Require evidence that 150 kW refers to the individual motor and identify whether it is peak or continuous	 CONFIRMED

1.5 Status definitions

Status tag	Meaning	Status
 CONFIRMED	Independently supported under a materially comparable boundary; numerical findings require two or three independent sources	 CONFIRMED
 CONTRADICTION	Eligible sources or the original report materially disagree, and the difference cannot be explained fully by boundary control	 CONTRADICTION
 ADJUSTED	Direction is supported, but the number, interpretation, attribution or boundary must be changed	 ADJUSTED
 GAP	Public evidence is insufficient to support a numerical value under a comparable boundary	 GAP

2. Evidence–base integrity and principal findings

2.1 Overall verdict

 ADJUSTED — The original report is strongest on qualitative architecture direction and weakest on 150 kW–normalized bills of material. Public evidence supports PMSM importance, rare–earth supply exposure, production use of EESM, specialized axial–flux commercialization and the central importance of thermal boundaries, but it does not validate most component–level Min | Mode | Max triplets [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#) [\[7\]](#).

 GAP — No supplied independent evidence provides a complete, independently weighed, motor–only bill of material for a representative 150 kW radial IPM/PMSM with matched peak or continuous rating. Consequently, the original stator, copper, rotor, magnet, shaft, housing, bearing, cooling and sensor masses cannot be called confirmed [\[2\]](#) [\[1\]](#).

 ADJUSTED — RAWCLIC’s torque–regression method is appropriate for scenario modelling only if the input component observations and vehicle–level motor–count boundary are visible. A regression confidence interval quantifies uncertainty around the fitted relationship; it does not correct source–boundary errors or convert sparse product observations into directly measured 150 kW bills of material [\[2\]](#).

2.2 Evidence streams retained

Evidence stream	Audit use	Principal eligible source base	Status
Rare–earth demand and motor dependence	Direction, supply concentration and substitution pressure	IEA, JRC and EU sources [3] [8] [4] [9] [10] [11] [12] [13] [14] [5]	 CONFIRMED
Production topology			

Evidence stream	Audit use	Principal eligible source base	Status
	OEM confirmation of EESM, axial-flux and integrated drive strategies	Renault, BMW, Mercedes-Benz and supplier sources [15] [16] [17] [18] [19] [20] [21] [22] [23] [24]	 CONFIRMED
Motor research and benchmarking	Topology, thermal limits, recycling and design trade-offs	Peer-reviewed and national-laboratory sources [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36]	 CONFIRMED
Axial-flux performance	Product specifications plus independent technical literature	YASA, IEEE and peer-reviewed sources [37] [38] [39] [40] [41] [42] [43] [44] [45] [46]	 ADJUSTED
Circularity and secondary magnets	Policy obligations, recycling routes and life-cycle effects	EU, JRC and peer-reviewed sources [47] [48] [49] [50] [51] [52] [53] [54] [55] [56]	 CONFIRMED
Long-term architecture adoption	Direction only; scenario values require explicit labelling	IEA, IDTechEx and JRC [57] [3] [4] [5] [7] [58] [59] [60]	 ADJUSTED

2.3 Values that survive cross-validation

 CONFIRMED — NdFeB traction magnets contain Nd and Pr as principal magnet rare earths, with Dy or Tb potentially used to improve coercivity and high-temperature performance [\[34\]](#) [\[9\]](#) [\[61\]](#).

 CONFIRMED — EESM and induction architectures eliminate rare-earth permanent magnets from the rotor, although they do not eliminate copper, electrical steel, housing, insulation or thermal-system requirements [\[4\]](#) [\[5\]](#) [\[6\]](#).

 ADJUSTED — PM traction motors are dominant, but the original mutually exclusive 78% | 85% | 92% radial-IPM share is not directly supported. The eligible evidence supports PM motor share above 75% since 2015 and separate estimates around 85% or above 90%, depending on vehicle, geography and counting scope [\[57\]](#) [\[62\]](#) [\[3\]](#) [\[7\]](#).

 CONFIRMED — Axial-flux machines can provide a short axial package and high torque or power density, but performance depends on topology, cooling, mechanical containment and whether peak or continuous power is used [\[63\]](#) [\[64\]](#) [\[65\]](#) [\[66\]](#) [\[43\]](#).

 ADJUSTED — A YASA P400-class value around 160 kW and 24–28.2 kg supports a peak motor-only density of approximately 5.7–6.7 kW/kg, not the original generic 3 | 5 | 8 kW/kg “production-oriented” band and not a continuous-density conclusion [\[42\]](#) [\[67\]](#) [\[68\]](#).

 CONFIRMED — DeepDrive is independently identified through Technical University of Munich sources as a TUM spin-off [\[69\]](#) [\[70\]](#) [\[71\]](#) [\[72\]](#) [\[73\]](#) [\[74\]](#).

 **ADJUSTED** — DeepDrive’s motor is a dual-rotor radial-flux topology, with a stator between inner and outer rotors; it is not an axial-flux architecture. Quantitative savings in magnet mass, iron, cost or efficiency remain company-attributed unless independently tested [75] [76] [73] [77].

3. Architecture mix and adoption claims

3.1 Current market position

Architecture claim	Original Min	Original Mode	Original Max	Revised Min	Revised Mode	Revised Max	Status
Radial IPM/ PMSM current share	78%	85%	92%	—	—	—	? GAP
EESM/WRS current share	4%	7%	10%	—	—	—	? GAP
Induction current share	3%	6%	10%	—	—	—	? GAP
Axial-flux current share	0%	1%	2%	—	—	—	? GAP
PM-motor market direction	Not separately stated	Approximately 85% to above 90%	Not separately stated	>75%	approximately 85%	>90% under broader scope	 ADJUSTED

 **ADJUSTED** — Evidence supports continued PM-motor dominance, but “PM motor” is not synonymous with “radial IPM.” Axial-flux PM machines and PM-assisted reluctance machines may fall inside a PM total while being separated in the original architecture table [57] [4] [5] [7].

 **CONTRADICTION** — The original architecture table describes its typical column as approximately normalized, but 85% + 7% + 6% + 1% equals 99%, while the individual uncertainty ranges overlap and use different classification concepts. The larger issue is that axial flux describes flux orientation whereas PMSM describes excitation, so the categories are not naturally mutually exclusive [1] [63] [64].

3.2 2030–2040 scenarios

 **GAP** — The original 2030, 2035 and 2040 architecture shares are scenario construction rather than confirmed forecasts. The supplied analyst evidence supports continuing PM dominance

through 2035 and increased interest in magnet-free, axial-flux and in-wheel technologies, but it does not independently reproduce the report's topology-specific triplets [57] [78] [79] [7] [58] [59].

 **ADJUSTED** — The only defensible quantitative trajectory in the supplied material is that one analyst direction places magnet-free automotive motors at approximately 9% in 2023 and approximately 30% in 2035. This does not determine the separate EESM and induction shares and must not be reallocated between them without the underlying dataset [57] [7] [59].

Horizon	Original interpretation	Revised interpretation	Status
2025/ current	Four mutually exclusive topology shares	PM dominance supported; topology-specific installed shares unresolved	 ADJUSTED
2030	Near-term share forecast	Scenario only; direction supported, exact shares unconfirmed	 GAP
2035	EESM and axial-flux expansion	Diversification supported; allocation by topology unconfirmed	 ADJUSTED
2040	Radial IPM mode falls to 61%	Speculative scenario without independent topology forecast	 GAP

3.3 2050–2070 scenarios

 **GAP** — The original 2045 and 2050 architecture shares are not independently corroborated. No supplied eligible source provides a comparable global or European installed-motor forecast dividing radial IPM, EESM, induction and axial flux through 2050 [1] [57] [3].

 **GAP** — The evidence on high-temperature superconducting motors concerns specialized aerospace, marine or heavy-duty research and cryogenic-system challenges. It does not support assigning passenger-BEV market shares for 2050–2070 [80] [81] [82] [83] [84].

 **ADJUSTED** — The evidence-supported long-term statement is limited to likely diversification under rare-earth supply pressure, continuing improvement in PM technology, and uncertain commercialization of alternative topologies. Numerical passenger-car shares beyond 2040 remain speculative scenario inputs [3] [4] [5] [59].

4. Motor mass, material composition and power density

4.1 The 150 kW radial IPM/PMSM bill of material

Component	Original Min	Original Mode	Original Max	Independently revised Min	Mode	Max	Status
Stator laminations	18.0 kg	23.0 kg	28.0 kg	—	—	—	 GAP
Stator copper	8.0 kg	11.0 kg	14.0 kg	—	—	—	

Component	Original Min	Original Mode	Original Max	Independently revised Min	Mode	Max	Status
							? GAP
Rotor laminations	10.0 kg	14.0 kg	18.0 kg	—	—	—	? GAP
NdFeB magnets	1.0 kg	1.5 kg	2.0 kg	—	—	—	? GAP
Shaft	3.0 kg	4.5 kg	6.0 kg	—	—	—	? GAP
Housing and end shields	8.0 kg	12.0 kg	17.0 kg	—	—	—	? GAP
Bearings and seals	1.0 kg	1.7 kg	2.5 kg	—	—	—	? GAP
Cooling hardware and fluid	2.0 kg	4.0 kg	7.0 kg	—	—	—	? GAP
Sensors and resolver	0.3 kg	0.7 kg	1.2 kg	—	—	—	? GAP
Total motor weight	55 kg	72 kg	90 kg	—	—	—	? GAP
Motor power density	1.7 kW/kg	2.1 kW/kg	2.7 kW/kg	—	—	—	? GAP

? GAP — Peer-reviewed sources confirm the physical material structure—electrical-steel stator and rotor cores, copper windings, permanent magnets where applicable, shaft, housing, insulation and bearings—but do not validate the original component masses for a matched 150 kW production motor [25] [27] [28] [29] [85] [86] [87].

🔧 ADJUSTED — The original report's 55 | 72 | 90 kg total corresponds arithmetically to 2.73 | 2.08 | 1.67 kW/kg at 150 kW, but this relationship does not validate either the mass or the power rating. The reported order of 1.7 | 2.1 | 2.7 kW/kg reverses the associated mass order and should be treated as a separately constructed range [1].

⚠️ CONTRADICTION — Recognised engineering evidence places contemporary motor-level power-density benchmarks materially above 1.7–2.7 kW/kg in some production and demonstrator contexts, while integrated-system values can be much lower. The disagreement cannot be resolved without knowing whether the original 150 kW is continuous, peak or vehicle-system power [88] [89] [90].

4.2 Magnet allowance and elemental composition

 **ADJUSTED** — Independent evidence supports a broad traction-motor magnet range of approximately 0.5–3 kg, but not a power-normalized 1.0 | 1.5 | 2.0 kg range specifically for a 150 kW motor. The original value is plausible as a planning band but remains unconfirmed [\[9\]](#) [\[91\]](#) [\[92\]](#).

Magnet parameter	Original Min	Original Mode	Original Max	Revised result	Status
NdFeB in a 150 kW IPM	1.0 kg	1.5 kg	2.0 kg	No matched 150 kW triplet; broad traction evidence approximately 0.5–3 kg	 ADJUSTED
Nd in motor	0.25 kg	0.38 kg	0.55 kg	No verified 150 kW elemental BOM	 GAP
Pr in motor	0.03 kg	0.08 kg	0.15 kg	No verified 150 kW elemental BOM	 GAP
Dy in motor	0.01 kg	0.05 kg	0.12 kg	No verified 150 kW elemental BOM	 GAP
Tb in motor	0.00 kg	0.01 kg	0.03 kg	No verified 150 kW elemental BOM	 GAP

 **CONFIRMED** — Published magnet evidence supports NdFeB compositions containing approximately 29–32% Nd and approximately 1–2% boron, with iron forming most of the balance and optional Dy, Tb or other additions. This chemistry does not determine motor-level elemental masses until magnet mass and grade are known [\[34\]](#) [\[61\]](#) [\[49\]](#).

 **ADJUSTED** — The original Dy range of 0.5% | 3% | 8% and Tb range of 0% | 0.5% | 2% are better interpreted as broad magnet-grade chemistry assumptions than as a confirmed representative 2025 traction-magnet distribution [\[34\]](#) [\[36\]](#) [\[61\]](#).

 **ADJUSTED** — Grain-boundary diffusion can reduce heavy-rare-earth use by concentrating Dy or Tb near grain boundaries, but the supplied evidence supports reductions around 50% or a broader 50–70% process range—not an independently validated 2030 industry-wide 50% | 60% | 70% outcome [\[34\]](#) [\[93\]](#) [\[94\]](#).

4.3 Scaling from 75 to 400 kW

 **GAP** — The original motor-mass and magnet-mass rows for 75, 100, 200, 250, 350 and 400 kW are not supported by matched public measurements. Motor mass does not scale uniquely with rated power because speed, torque, duration, cooling, topology and integration change simultaneously [\[1\]](#) [\[6\]](#) [\[95\]](#).

 **CONTRADICTION** — The original report cautions that power does not scale linearly with mass, yet provides a complete ladder of smooth mass and magnet triplets without a disclosed sizing model or matched empirical dataset. The caution is valid; the numerical ladder is not independently reproducible [\[1\]](#).

5. EESM, induction, ferrite and synchronous–reluctance alternatives

5.1 EESM comparison

✓ CONFIRMED — Renault and BMW public material confirms that current–excited or electrically excited synchronous traction machines are production strategies rather than purely academic alternatives [15] [16] [18] [19] [20].

✓ CONFIRMED — EESM eliminates the traction permanent magnet but adds rotor excitation, rotor conductors and associated control and thermal requirements [29] [30] [31] [5] [6].

EESM parameter relative to IPM	Original Min	Original Mode	Original Max	Revised value	Status
Motor–mass difference	+4 kg	+8 kg	+15 kg	No comparable public triplet	? GAP
Additional rotor copper	3 kg	6 kg	10 kg	No comparable public triplet	? GAP
Highway efficiency difference	–1.5 pp	–0.5 pp	+0.5 pp	Direction depends on operating point; no universal triplet	? GAP
NdFeB reduction	1.0 kg	1.5 kg	2.0 kg	Magnet eliminated relative to the selected IPM baseline; absolute amount unresolved	 ADJUSTED

⚠ CONTRADICTION — Comparative studies do not establish a universal highway–speed efficiency penalty for EESM. Results vary with motor optimization, field excitation, speed, load and cooling, so a single cross–vehicle triplet can cross zero without providing a decision–useful central estimate [29] [30] [96] [97].

5.2 Induction machines

✓ CONFIRMED — Induction motors avoid permanent magnets and therefore avoid Nd, Pr, Dy and Tb in the rotor magnet system, while retaining substantial electrical–steel, conductor and cooling requirements [4] [5] [95].

? GAP — The original current and future induction shares are not confirmed because evidence mixes primary traction motors, secondary axle motors, vehicle models and installed motors [57] [7] [59].

5.3 Ferrite and synchronous–reluctance alternatives

✓ CONFIRMED — Ferrite–assisted synchronous–reluctance machines are an active rare–earth–free or rare–earth–reduced research route. Ferrite can improve power factor and torque relative to

a pure synchronous–reluctance rotor, but its lower magnetic performance requires topology and volume compensation [27] [87] [98] [99].

 **CONFIRMED** — The ReFreeDrive evidence includes a 200 kW peak synchronous–reluctance design and a 75 kW peak scaled version, demonstrating technical development at traction–relevant power rather than proving mass–market adoption [100] [101].

 **GAP** — No supplied evidence supports a 150 kW ferrite–assisted motor–only BOM, magnet mass, continuous power density or production–market share. A numeric substitution scenario would therefore be manufactured.

6. Axial flux, YASA, DeepDrive and in–wheel systems

6.1 Axial–flux weight and density claims

Axial–flux parameter	Original Min	Original Mode	Original Max	Revised Min	Revised Mode	Revised Max	Status
Weight saving versus radial motor	10%	20%	30%	—	—	—	 GAP
Saving at 150 kW	7 kg	14 kg	24 kg	—	—	—	 GAP
“Production–oriented” density	3 kW/kg	5 kW/kg	8 kW/kg	—	—	—	 GAP
YASA P400–class peak density	Not separated	Not separated	Not separated	5.7 kW/kg	approximately 6.0 kW/kg	6.7 kW/kg	 ADJUSTED

 **ADJUSTED** — Academic reviews support a directional axial–flux advantage in torque density and axial package length, but do not validate a universal 10% | 20% | 30% mass saving against an unspecified radial comparator [63] [64] [65] [66].

 **ADJUSTED** — A 7 | 14 | 24 kg saving is derived from applying percentage assumptions to the unconfirmed 55 | 72 | 90 kg radial baseline. It is therefore a second–order scenario calculation, not independent product evidence [1].

6.2 YASA continuous–versus–peak caveat

 **CONFIRMED** — The YASA P400 family is an axial–flux, permanent–magnet, yokeless and segmented–armature topology [38] [42] [68].

🔧 ADJUSTED — Public specifications support approximately 160 kW peak and dry mass around 24–28.2 kg for P400 variants, but continuous output varies substantially by version and cooling configuration. The peak ratio must not be described as continuous power density [102] [67] [68].

⚠️ CONTRADICTION — The original report presents 3 | 5 | 8 kW/kg as a production-oriented axial-flux range without tying it to duty duration. The P400 evidence demonstrates why this is unsafe: approximately 160 kW peak divided by variant mass produces a high peak density while cited continuous outputs span approximately 20–100 kW [103] [67] [68].

🔧 ADJUSTED — Later YASA prototype records remain company-attributed short-duration claims unless independent test data establish rating duration, coolant conditions, inverter boundary and repeatability [37] [104] [39] [40].

6.3 DeepDrive verification

✅ CONFIRMED — TUM sources identify DeepDrive as a Technical University of Munich spin-off [69] [70] [71] [72] [73] [74].

✅ CONFIRMED — Independent technical and company descriptions identify the core topology as a dual-rotor radial-flux machine with a stator located between inner and outer rotors [75] [76] [73] [77].

🔧 ADJUSTED — Claims of approximately 50% less magnet material, 80% less iron, broad efficiency above 96% or cost reductions above 70% remain company-attributed in the supplied evidence. Media repetition does not constitute independent experimental validation [105] [75] [76] [77].

🔧 ADJUSTED — DeepDrive offers central-drive and in-wheel development routes, but its 150 kW, 24 kg MG 250 is described as a motor-generator for a range extender. It is not evidence for a 150 kW BEV traction-motor BOM [106].

6.4 In-wheel systems

✅ CONFIRMED — In-wheel motors can remove conventional central driveline elements and enable direct wheel control, but they create distinct unsprung-mass, sealing, impact, braking, thermal and service boundaries [57] [107] [108] [109].

❓ GAP — No supplied source supports an installed passenger-BEV in-wheel architecture share for 2025, 2030, 2040 or later horizons.

🔧 ADJUSTED — DeepDrive and Continental development activity demonstrates technical and OEM interest, not confirmed high-volume adoption [110] [76] [77].

7. Circularity, life, scrap and EU policy

7.1 Recycled and secondary NdFeB

✓ CONFIRMED — The EU Critical Raw Materials Act is Regulation (EU) 2024/1252 and establishes a 2030 benchmark under which at least 25% of annual EU consumption of strategic raw materials should come from EU recycling capacity [\[47\]](#) [\[48\]](#) [\[55\]](#) [\[111\]](#).

🔧 ADJUSTED — The 25% CRMA benchmark applies to strategic raw materials collectively; it is not a guarantee that 25% of every traction motor's NdFeB magnet mass will be recycled content in 2030 [\[47\]](#) [\[48\]](#) [\[55\]](#).

✓ CONFIRMED — The CRMA introduces information obligations concerning permanent magnets, including presence, recyclability and recycled-content information for covered products [\[56\]](#) [\[112\]](#) [\[113\]](#) [\[114\]](#) [\[115\]](#).

✓ CONFIRMED — EU and JRC evidence describes current rare-earth permanent-magnet recycling as below 1%, establishing a low starting point for secondary supply [\[12\]](#) [\[116\]](#) [\[117\]](#) [\[118\]](#).

🔧 ADJUSTED — Recycling can reduce primary-material dependence, but the contribution of end-of-life EV motors remains constrained by vehicle lifetime, collection, dismantling, magnet identification, feedstock quality and processing capacity [\[50\]](#) [\[51\]](#) [\[53\]](#) [\[119\]](#) [\[120\]](#).

7.2 Magnet manufacturing scrap

✓ CONFIRMED — NdFeB manufacturing generates clean scrap, machining swarf and sludge that are generally easier to collect than dispersed end-of-life magnets [\[121\]](#) [\[91\]](#) [\[52\]](#) [\[53\]](#).

🔧 ADJUSTED — Evidence indicates manufacturing scrap can represent approximately 20–30% of raw-material input in some sintered-magnet processes. This is a process-input or gross-scrap range, not necessarily unrecovered net waste [\[121\]](#) [\[91\]](#) [\[53\]](#).

✓ CONFIRMED — Hydrogen processing is suited particularly to clean magnet scrap, while hydrometallurgical routes can process oxidized or contaminated feedstocks and may achieve high rare-earth recovery under specified laboratory or process conditions [\[50\]](#) [\[51\]](#) [\[52\]](#) [\[53\]](#) [\[122\]](#).

? GAP — The original report contains no explicit manufacturing-yield or recycled-swarf assumption in its 150 kW material table. Gross motor BOM, purchased magnet input and net virgin-material demand therefore cannot be reconciled.

7.3 Motor life and end-of-life flows

✓ CONFIRMED — Traction-motor life depends on thermal cycling, winding-insulation degradation, bearings, lubrication, electrical stresses, rotor integrity and, for PM motors, demagnetization risk [\[123\]](#) [\[124\]](#) [\[125\]](#) [\[126\]](#) [\[127\]](#).

✓ CONFIRMED — Reuse, remanufacturing and recycling are distinct end-of-life routes; functional disassembly and condition diagnosis determine whether material recycling is preferable to retaining component value [128] [129] [130] [131] [132].

? GAP — No supplied evidence provides a representative passenger-BEV traction-motor lifetime distribution by topology, vehicle segment, climate or duty cycle. A single lifetime in years or kilometres would be unsupported.

? GAP — No supplied evidence provides EU annual retirement flows of PM, induction and EESM motors with recoverable magnet, copper, steel and aluminium masses. Stock-and-flow modelling therefore requires sensitivity cases rather than a confirmed end-of-life curve.

7.4 Environmental interpretation

✓ CONFIRMED — Life-cycle evidence identifies NdFeB production as an important environmental and supply-risk contributor, while copper, electrical steel and aluminium also contribute materially to traction-motor impacts [65] [133] [134] [49] [135].

🔧 ADJUSTED — Magnet-to-magnet recycling can reduce impacts relative to virgin magnet production, but reported reductions depend on feedstock, process, electricity and allocation boundaries. They should not be applied directly to the entire motor mass [65] [134] [50].

8. Vehicle configuration, e-axle boundaries, thermal duty and regulation

8.1 Single- versus dual-motor vehicles

✓ CONFIRMED — RAWCLIC reports component masses at vehicle level, which is suitable for material-flow accounting only if the number and type of installed motors are represented correctly [2].

⚠ CONTRADICTION — The original practical report moves between motor-level 150 kW values, vehicle-level dual-motor applications and higher-power classes without a consistent installed-motor boundary. A 300 kW dual-motor vehicle is not materially equivalent to one 300 kW motor [1].

? GAP — Eligible public evidence in the supplied data does not provide an EU registration-weighted distribution of single-, dual- and multi-motor BEVs by segment and year.

🔧 ADJUSTED — Material-demand modelling should use installed motors per vehicle, allocating front and rear machines separately where topology or rating differs. Vehicle nameplate power alone is insufficient.

8.2 E-axle and drive-unit boundaries

✓ CONFIRMED — Modern electric drive systems frequently integrate the motor, inverter, reduction gear, differential and cooling functions. Published drive-unit mass must therefore not be assigned to the bare motor without disassembly evidence [6] [136] [137] [123].

⚠ CONTRADICTION — The original report correctly warns against using e-axle masses as motor masses, yet its 55 | 72 | 90 kg motor range is said to be anchored partly by product and drive-unit evidence. That evidence cannot confirm the motor-only range unless the non-motor content is removed [1].

? GAP — The supplied evidence does not provide separable masses for Volkswagen APP550, Hyundai E-GMP, BYD e-Platform 3.0 or the Tesla Model 3 rear motor under one consistent boundary.

8.3 Thermal continuous-versus-peak duty

✓ CONFIRMED — Continuous output is constrained principally by losses, heat transfer and allowable temperatures, whereas peak output can be sustained only for a specified duration and initial thermal condition [6] [95] [138] [123].

⚠ CONTRADICTION — Any density claim using an unspecified “150 kW” numerator is indeterminate. The original radial and axial density tables do not consistently identify whether 150 kW is peak or continuous [1].

🔧 ADJUSTED — Cooling hardware cannot be treated merely as parasitic mass. More capable cooling may increase motor assembly mass while enabling a much higher continuous rating, thereby improving continuous power density at system level [6] [95] [138].

? GAP — No original mass scenario specifies coolant inlet temperature, flow rate, maximum winding temperature, magnet temperature, duty duration or steady-state criterion. The 2030–2040 mass reductions therefore lack a thermal acceptance basis.

8.4 Euro 7 and passenger-car power limits

✓ CONFIRMED — Euro 7 introduces requirements addressing non-exhaust emissions, including brake-particle emissions, alongside broader vehicle-emissions and durability provisions [139] [140] [141] [142] [143] [144].

⚠ CONTRADICTION — No supplied EU legislation establishes a general passenger-car traction-motor power limit in kW, and the original evidence does not justify one [139] [140] [141] [142].

⚠ CONTRADICTION — Brake-particle limits do not establish a causal regulatory rule reducing or capping BEV motor power. Motor kW must not be projected from Euro 7 brake-particle provisions [139] [140] [141] [143].

🔧 ADJUSTED — Regenerative braking may change friction-brake use, but any effect on brake particles depends on vehicle control, battery acceptance, brake blending, brake design and driving conditions. The supplied evidence does not quantify a motor–power relationship.

9. Dedicated contradictions and gaps

9.1 Contradictions

C-1 — Architecture classification

⚠️ CONTRADICTION — The original report treats radial IPM/PMSM and axial flux as mutually exclusive primary architectures [1], while axial flux describes flux orientation and can also use permanent-magnet synchronous excitation [63] [64]. Resolution: use two dimensions—flux orientation and excitation type—rather than one exclusive list.

C-2 — PMSM share versus radial-IPM share

⚠️ CONTRADICTION — IEA and analyst evidence supports high PM-motor penetration [57] [3] [7], but the original report converts that evidence into a radial-IPM share of 78% | 85% | 92% [1]. Resolution: retain PM dominance as directional evidence; withdraw the radial-IPM triplet.

C-3 — Motor-only mass versus integrated-drive evidence

⚠️ CONTRADICTION — The original report warns that integrated drive-unit masses cannot be treated as motor masses but uses drive-unit evidence to anchor a 55 | 72 | 90 kg motor-only envelope [1]. Fraunhofer evidence confirms that motor, inverter and gearbox integration is widespread [6] [136]. Resolution: mark the 150 kW motor-only mass as a gap.

C-4 — Radial power-density benchmark

⚠️ CONTRADICTION — The original 1.7 | 2.1 | 2.7 kW/kg radial range [1] is materially below some published motor-level production benchmarks and above some integrated-system values [88] [89] [90]. Resolution: no revised triplet until rating duration and denominator are matched.

C-5 — YASA peak and continuous density

⚠️ CONTRADICTION — The original report presents a generic 3 | 5 | 8 kW/kg axial-flux range [1], while YASA P400 evidence distinguishes approximately 160 kW peak from much lower and configuration-dependent continuous power [103] [67] [68]. Resolution: report peak and continuous densities separately.

C-6 — EESM efficiency penalty

⚠️ CONTRADICTION — The original -1.5 | -0.5 | +0.5 percentage-point highway difference [1] implies a transferable comparison, while peer-reviewed comparisons vary by operating point and optimization [29] [30] [96] [97]. Resolution: retain qualitative duty-cycle dependence and withdraw the universal triplet.

C-7 — Future shares represented numerically

⚠️ CONTRADICTION — The original report calls the 2030–2050 values scenarios but presents complete topology tables with forecast-like precision [1]. Analyst evidence supports diversification without confirming those allocations [57] [7] [59]. Resolution: separate evidence-supported direction from explicit scenario construction.

C-8 — Scaling warning versus scaling table

⚠️ CONTRADICTION — The report states that power does not scale linearly with mass, yet supplies smooth 75–400 kW mass and magnet ladders without a disclosed physical model [1]. Resolution: remove the ladder from confirmed-value use.

C-9 — DeepDrive topology

⚠️ CONTRADICTION — Secondary descriptions can associate DeepDrive with wheel-integrated concepts, but TUM, technical and company evidence identifies a dual-rotor radial-flux machine rather than an axial-flux machine [75] [76] [73] [77]. Resolution: classify it as dual-rotor radial flux.

C-10 — Euro 7 and motor power

⚠️ CONTRADICTION — Euro 7 brake-particle provisions are regulatory emissions requirements [139] [140] [141], not an EU passenger-car traction-power cap. Resolution: exclude any motor-kW limit or causal forecast derived from brake-particle rules.

9.2 Gaps

G-1 — Recycled and secondary NdFeB

❓ GAP — No evidence quantifies recycled NdFeB content in a representative 150 kW motor by 2025, 2030 or 2040. Fill method: obtain supplier declarations using CRMA-compatible recycled-content methodology and distinguish pre-consumer magnet scrap from post-consumer end-of-life material [55] [56] [114].

G-2 — Motor life and end-of-life

❓ GAP — No topology-specific survival curve is available. Fill method: combine warranty returns, dismantler observations, odometer-at-retirement data and thermal-duty models, then validate separate reuse, remanufacture and recycling probabilities [125] [126] [130].

G-3 — Ferrite alternatives

❓ GAP — No 150 kW ferrite-assisted production BOM or continuous density is disclosed. Fill method: require a matched IPM/ferrite demonstrator comparison using the same stator outer diameter, cooling, voltage, peak duration and continuous thermal limit [98] [99].

G-4 — Manufacturing scrap

 GAP — The gross 20–30% magnet–process scrap range does not reveal internal recycling, external recycling or net virgin–material loss. Fill method: collect mass–balance data from alloy input through sintering, machining, rejected magnets and recovered swarf [121] [91] [53].

G-5 — Dual- versus single-motor vehicles

 GAP — Registration data do not identify installed motor count and topology consistently. Fill method: map vehicle variants and axle configurations to registrations, retaining separate front and rear motor records [2].

G-6 — In-wheel adoption

 GAP — Demonstrators and supplier programmes do not establish market penetration. Fill method: count homologated series–production vehicles and installed wheel motors rather than announcements, prototypes or funding rounds [57] [77].

G-7 — E-axle boundary

 GAP — Public product specifications rarely disaggregate motor, inverter, gearbox, differential, oil and housing masses. Fill method: use controlled teardowns with a boundary protocol and photographically traceable weighing of each cleaned subassembly [6] [136].

G-8 — Thermal peak and continuous duty

 GAP — Original density values lack duration and coolant conditions. Fill method: report at least peak power and duration, continuous steady–state power, coolant inlet temperature, flow rate, winding limit, magnet limit and included cooling mass [6] [95] [138].

10. Master confirmed–value table

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
Radial IPM/ PMSM	Current architecture share	78% 85% 92%	—	—	—	 GAP
PM motors, all relevant excitation/ orientation classes	Current market direction	Approximately 85% to >90%	>75%	approximately 85%	>90% under broader scope	 ADJUSTED
EESM/WRSM		4% 7% 10%	—	—	—	 GAP

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
	Current architecture share					
Induction	Current architecture share	3% 6% 10%	—	—	—	? GAP
Axial flux	Current architecture share	0% 1% 2%	—	—	—	? GAP
Radial IPM/ PMSM, 150 kW class	Total motor mass	55 72 90 kg	—	—	—	? GAP
Radial IPM/ PMSM, 150 kW class	Power density	1.7 2.1 2.7 kW/kg	—	—	—	? GAP
Radial IPM/ PMSM, 150 kW class	Stator laminations	18 23 28 kg	—	—	—	? GAP
Radial IPM/ PMSM, 150 kW class	Stator copper	8 11 14 kg	—	—	—	? GAP
Radial IPM/ PMSM, 150 kW class	Rotor laminations	10 14 18 kg	—	—	—	? GAP

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
Radial IPM/ PMSM, 150 kW class	NdFeB magnets	1.0 1.5 2.0 kg	0.5 kg broad traction range	—	3.0 kg broad traction range	 ADJUSTED
Radial IPM/ PMSM, 150 kW class	Shaft	3.0 4.5 6.0 kg	—	—	—	 GAP
Radial IPM/ PMSM, 150 kW class	Housing and end shields	8 12 17 kg	—	—	—	 GAP
Radial IPM/ PMSM, 150 kW class	Bearings and seals	1.0 1.7 2.5 kg	—	—	—	 GAP
Radial IPM/ PMSM, 150 kW class	Cooling hardware and fluid	2 4 7 kg	—	—	—	 GAP
Radial IPM/ PMSM, 150 kW class	Sensors and resolver	0.3 0.7 1.2 kg	—	—	—	 GAP
NdFeB magnet	Nd content	Motor-level 0.25 0.38 0.55 kg	29% of magnet mass	approximately 30%	32%	 ADJUSTED
NdFeB magnet	Dy content	0.5% 3% 8%	—	—	—	 GAP
NdFeB magnet	Tb content	0% 0.5% 2%	—	—	—	 GAP

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
NdFeB magnet	GBD heavy-rare-earth reduction	50% 60% 70% by 2030	approximately 50%	—	approximately 70% process range	 ADJUSTED
EESM	Motor-mass difference versus IPM	+4 +8 +15 kg	—	—	—	 GAP
EESM	Additional rotor copper	3 6 10 kg	—	—	—	 GAP
EESM	Highway efficiency difference	-1.5 -0.5 +0.5 pp	—	—	—	 GAP
EESM	Magnet elimination	1.0 1.5 2.0 kg reduction	Full elimination of selected IPM magnet mass	Same	Same	 ADJUSTED
Ferrite PMa-SynRM	150 kW BOM and density	Not provided	—	—	—	 GAP
Pure SynRM	Demonstrated traction power class	Not provided	75 kW peak scaled design	—	200 kW peak design	 ADJUSTED

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
Axial-flux PM	Weight saving versus radial	10% 20% 30%	—	—	—	? GAP
Axial-flux PM	Weight saving at 150 kW	7 14 24 kg	—	—	—	? GAP
Axial-flux PM	Generic production density	3 5 8 kW/kg	—	—	—	? GAP
YASA P400-class axial flux	Peak power	Approximately 160 kW	160 kW	160 kW	160 kW	✓ CONFIRMED
YASA P400-class axial flux	Dry motor mass	Approximately 24 kg	24 kg	—	28.2 kg	🔧 ADJUSTED
YASA P400-class axial flux	Peak power density	Not separated	5.7 kW/kg	approximately 6.0 kW/kg	6.7 kW/kg	🔧 ADJUSTED
YASA P400-class axial flux	Continuous power	Not separated	approximately 20 kW	—	approximately 100 kW	🔧 ADJUSTED
DeepDrive dual-rotor radial flux	TUM spin-off identity	Not explicitly verified	Yes	Yes	Yes	✓ CONFIRMED
Topology		Dual rotor		Radial flux		✓ CONFIRMED

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
DeepDrive dual-rotor radial flux		Ambiguous in secondary discussion		Stator between rotors		
DeepDrive dual-rotor radial flux	Magnet saving	Approximately 50% claim	—	—	—	 ADJUSTED
DeepDrive dual-rotor radial flux	Iron saving	Approximately 80% claim	—	—	—	 ADJUSTED
DeepDrive MG 250	150 kW and 24 kg	Potentially comparable 150 kW evidence	—	—	—	 ADJUSTED
All architectures	2030 topology shares	Complete original triplets	—	—	—	 GAP
All architectures	2035 topology shares	Complete original triplets	—	—	—	 GAP
All architectures	2040 topology shares	Complete original triplets	—	—	—	 GAP
All architectures	2050 topology shares	Complete original triplets	—	—	—	 GAP

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
Passenger-BEV motors	EU power limit	Implicit policy concern	—	—	—	 CONTRADICTION
Passenger-BEV motors	Brake-particle rule causes lower motor kW	Potential causal inference	—	—	—	 CONTRADICTION
Strategic raw materials	EU recycling benchmark for 2030	Not integrated into motor scenarios	25%	25%	25%	 CONFIRMED
NdFeB magnets	2030 motor-level recycled content	Not provided	—	—	—	 GAP
NdFeB manufacture	Gross scrap or swarf generation	Not provided	20%	—	30%	 ADJUSTED
Traction motors	Service-life distribution	Not provided	—	—	—	 GAP
Traction motors	End-of-life route shares	Not provided	—	—	—	 GAP

Motor type	Parameter	Original value	Confirmed Min	Confirmed Mode	Confirmed Max	Status
Vehicle-level propulsion	Single- versus dual-motor distribution	Not consistently controlled	—	—	—	? GAP
In-wheel motors	Passenger-BEV adoption share	Not provided	—	—	—	? GAP
Integrated e-axles	Motor-only mass disaggregation	Treated cautiously but used as anchor	—	—	—	? GAP
All motor types	Continuous/peak density conversion	Often unspecified	—	—	—	? GAP

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